$$y = (x-a)^n e^{bx}\sqrt{1+x^2}$$

$$\begin{aligned}\frac{dy}{dx} &= n(x-a)^{n-1}e^{bx}\sqrt{1+x^2} + b(x-a)^n e^{bx}\sqrt{1+x^2} + \frac{x(x-a)^n e^{bx}}{\sqrt{1+x^2}} \\ &= \frac{(x-a)^{n-1}e^{bx}}{\sqrt{1+x^2}}\left[n(1+x^2) + b(x-a)(1+x^2) + x(x-a)\right]\end{aligned}$$

$$q(x) = n(1+x^2) + b(x-a)(1+x^2) + x(x-a)$$

$$q(x) = bx^3 + (n+1-ab)x^2 + (b-a)x + n - ab$$

$\because b \neq 0,\ q(x)$ is cubic.

(i) When $a = 4,\ b = 4,\ n = 15$,

$$q(x) = 4x^3 + (16-16)x^2 + (4-4)x + 15 - 16 = 4x^3 - 1$$

$$\int \frac{(x-4)^{14}e^{4x}(4x^3-1)}{\sqrt{1+x^2}}\,dx = (x-4)^{15}e^{4x}\sqrt{1+x^2} + C$$

(ii)

$$12x^4 - x^2 - 11 = (x-1)(12x^3 + 12x^2 + 11x + 11)$$

When $a = 1,\ b = 12,\ n = 23$,

$$q(x) = 12x^3 + 12x^2 + 11x + 11$$

$$\begin{aligned}\int \frac{(x-1)^{21}e^{12x}(12x^4 - x^2 - 11)}{\sqrt{1+x^2}}\,dx &= \int \frac{(x-1)^{22}e^{12x}q(x)}{\sqrt{1+x^2}}\,dx \\ &= (x-1)^{23}e^{12x}\sqrt{1+x^2} + C\end{aligned}$$

(iii) Let $a = 2,\ b = 4,\ n = 7$,

$$q(x) = 4x^3 + 2x - 1$$

$$\frac{d}{dx}\left[(x-2)^7 e^{4x}\sqrt{1+x^2}\right] = \frac{(x-2)^6 e^{4x}(4x^3+2x-1)}{\sqrt{1+x^2}}$$

$$
\begin{aligned}
&\frac{d}{dx}\left[x(x-2)^7 e^{4x}\sqrt{1+x^2}\right] \\
&= x \cdot \frac{(x-2)^6 e^{4x}(4x^3+2x-1)}{\sqrt{1+x^2}} + (x-2)^7 e^{4x}\sqrt{1+x^2} \\
&= \frac{(x-2)^6 e^{4x}}{\sqrt{1+x^2}}\left[x(4x^3+2x-1) + (x-2)(1+x^2)\right] \\
&= \frac{(x-2)^6 e^{4x}(4x^4+x^3-2)}{\sqrt{1+x^2}}
\end{aligned}
$$

$$
\int \frac{(x-2)^6 e^{4x}(4x^4+x^3-2)}{\sqrt{1+x^2}}\,dx = x(x-2)^7 e^{4x}\sqrt{1+x^2} + C
$$

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